Document Overview

- This document is a living artifact.
- It will evolve as the Product Backlog evolves.
- Sprint 0 is not for building features.
- Complete each section according to the instructions.

Section 1 – Product Goal (Long-Term Objective)

- Your produce goal should be a clear, measurable statement describing the final state of the product.
- Requirements:
    - Must describe value.
    - Must be achievable by the end of the project timeline.
    - Must not describe tasks.
    - Must not describe activity.
    - Must describe outcome.
- Example format: By the end of this project, users will be able to _____ so that _____.
- Add your product goal in the space below.

By the end of this project, users will be able to use the camera to analyze the nutrition label of a product or scan the barcode, either using OCR or AI to parse the nutrition label text and ingredients. AI will be able to explain what long, complex, and convoluted ingredients are and if they are bad for you, alert users if there are any allergens, and recommend substitutes that are better and healthier.

Section 2 – Problem Statement

- What problem are we solving?
- Who experiences this problem?
- Why does it matter?
- What happens if this problem is not solved?

Write 1 – 2 strong paragraphs answering these questions in the space below.

Nutrition labels are legally required on all food products but inaccessible for the average person. Ingredients like "sodium benzoate," or "partially hydrogenated soybean oil" appear on millions of products, yet most consumers have no clue what they mean and the risks they pose.

Existing nutrition apps focus on counting calories and tracking macros instead of ingredient education. They show the numbers and percentages behind the food you eat, but do not explain how those numbers tie to real health outcomes. Consumers with allergies, dietary restrictions, or chronic conditions are the people who pay the price. This results in people either trusting labels blindly, avoiding packaged food entirely out of anxiety, or spending excessive time researching individual ingredients. None of these are satisfactory solutions, and none of them promote a long-term understanding of healthy food choices.

Section 3 – Target User & Primary Persona

- Identify and define your primary user persona (the one driving prioritization decisions)
    - Age/Context
    - Goals
    - Pain points
    - Technical ability
    - Why they would use this product
- Age/Context
    - Middle-aged, millennial, 28–45 years old. Typically shopping for a household/family.
- Goals
    - To identify ingredients (dyes, high-fructose corn syrup, processed additives) that negatively impact childhood development, health, etc.
- Pain points
    - Products that market themselves as healthy when they aren't actually (e.g. labeled natural when they actually contain synthetic ingredients). Requires too much time/effort to research ingredients in a grocery aisle.
- Technical ability
    - Moderate, use devices like phone/computer and apps like banking/social media on a daily basis.
- Why they would use this product
    - To get an instant, honest translation/summary of what a product actually contains. Helps users understand and eliminates confusion from nebulous marketing claims and copious amounts of raw data.
- Identify and define your secondary user persona (if relevant)
    - Age/Context
    - Goals
    - Pain points
    - Technical ability
    - Why they would use this product
- Age/Context
    - 18-35 years old, interested in living longer, living healthy, and staying fit and active.
- Goals
    - Find the healthiest version of products like protein bars and plant milks.
- Pain points

    - Existing apps focus on mainly calories and macronutrients like proteins, fats, and carbs, rather than ingredients.
- Technical ability
    - High technical ability, they should be able to use advanced features like toggling filters, viewing detailed breakdowns, etc.
- Why use this product
    - Verify the healthiness of products so that they don't hinder their fitness goals.
- If trade-offs occur, whose needs win?
- We will always prioritize our primary user persona. If a feature or piece of data becomes too technical or dense, we will pivot and change it so that it is simpler and more readable. For example, if a secondary user persona wants a detailed breakdown but it makes the screen too cluttered for a busy parent to read in 5 seconds, we will hide the data behind a "show more" button.

Section 4 – Minimum Viable Product (MVP)

Define clearly:

1. What must exist for MVP?
2. What will NOT be included in MVP?
3. How will we know MVP is complete?
4. What can be demonstrated at MVP stage?

MVP must:

- Be demonstrable
- Be valuable
- Be constrained

Perfection is not MVP

- What must exist for MVP?
    - A functioning camera-based barcode/nutrition label scanner that is connected to the Open Food Facts API

    - AI that takes in the list of ingredients and outputs a summary of the healthiness of the product
    - A recommendation system for a couple healthier alternatives
    - Warnings for allergens, unhealthy ingredients, etc.
- What will NOT be included in MVP?
    - OCR to scan the nutrition label's text directly
    - Social media features like community ratings
    - Option for comparing two or more products directly in terms of healthiness
    - Saving favorites and history of scanned items
    - Login/registration system
    - Data-packaging for food conglomerates
    - A rating system for the product's healthiness
- How will we know MVP is complete?
    - The time from reading the barcode to displaying the summary is under 5 seconds
    - The AI correctly identifies and explains at least 90% of ingredients for products in a standard grocery store aisle
    - The app can be given to a stranger who has never used it before and they are able to use it successfully without instructions
- What can be demonstrated at MVP stage?
    - Use the app to open the camera and scan a product and it'll spit out a health summary and point out harmful ingredients
    - An option to see healthier alternatives to the scanned product

Section 5 – Initial Product Backlog (Ordered)

Create an ordered list of 15-25 backlog items.

Each item must:

- Be written as a user story
- Be small enough to complete within a Sprint
- Be sized roughly (S/M/L)

Format: As a _____, I want _____ so that _____.

Highlight the top 5 highest priority items.

Important: This backlog will change.

Backlog Number	As a user…	Priority	Size	Sprint	Acceptance Criteria
1	Open the app and acceses camera scanner	High	Small	Sprint 1	Camera opens, UI visible, no crashes
2	Point camera at nutrition label and capture image	High	Medium	Sprint 1	Image is captured and passe to processing
3	Scan barcode and have app retrieve product information	High	Medium	Sprint 1	Barcode decoded, basic info returned
4	Plain-english breakdown of nutrition label values	High	Medium	Sprint 1	All standard label fields displayed clearly with context
5	Receive an alert if scanned product contains any selected allergens	High	Small	Sprint 1	Alert notified, no false negatives on common allergies
6	Configure allergen profile	High	Small	Sprint 1	Select allergens, preferences saved across sessions
7	See explanation of complex or unfamiliar ingredients	High	Large	Sprint 2	Ingredient flagged as complex gets description and context

8	Warned if an ingredient is associated with health concerns	High	Medium	Sprint 2	Concerning ingredients highlighted with explanation that is credible
9	See healthier substitute product recommendations	High	Large	Sprint 2	At least 1-3 alternatives shown with comparison of nutrient metrics
10	Log a scanned product to daily nutrition diary	High	Medium	Sprint 2	Produced added to daily log and total nutrients calculated correctly
11	View a summary of daily nutrient intake	Medium	Medium	Sprint 2	Dashboard shows calories, protein, fat, carbs vs recommended daily values for stats
12	View scan history from previous days	Medium	Small	Sprint 3	The history list shows past scans with date, name, and summary.
13	Search for a product by name if the scan fails.	Medium	Medium	Sprint 3	Search returns relevant products, and users can see full label breakdown
14	Create an account and log in to sync data across devices	Medium	Large	Sprint 3	Register/login with email, data contained after logout and re-login
15	Share product scan result or health tip with others	Small	Small	Sprint 4	Share sheet opens, content formatted correctly for social apps
16	Set weekly nutrition goals and see progress	Small	Medium	Sprint 4	Goal input saved, progress charts updated based on logged items
17	Daily summary notification with nutrition score	Small	Medium	Sprint 4	Push notification delivered daily, tapping it opens summary
18	Ability to group together items and view nutrients in cooked meals	Small	Medium	Sprint 4	Can compile together multiple barcodes, creating meals that show the nutrients from that meal

Section 6 – Definition of Done (Team Agreement)

Work is only considered complete when:

- It works without errors.
- It can be demonstrated without explanation.
- It is merged into the main branch.
- It is peer reviewed
- It meets team quality standards.

Discuss, agree on, and explicitly write out your teams:

- Code standards (including style): No magic numbers in code, properly named variables, informative comments, correct naming conventions, non-redundant code, modular code.
- Testing expectations: Test every possible combination of user inputs. Ensure we test edge cases on all our features, ensuring no mishap occurs during demos.
- Review requirements: Ensure that all the requirements are met, go back and forth between requirements and complete a checklist all together before demonstrating.
- Merge policy: Testing merge requests content completely with proper review of requirements and group Sprint Goal. Person merging should complete comments that inform the user of their work clearly. Person reviewing the merger should provide comments explaining both the good and bad of the merger request. Merges should be done regularly, to keep our product constantly developing.
- Demo expectations: All members present an equal amount and make sure to thoroughly go through all of the requirements and all of the features. All main user functions should be displayed and presented in a well-rehearsed manner. All members should understand the complete demo and be able to answer questions about any part of the iteration.

All team members must sign this section.

Arda

Siddharth

Varun

Leo

Section 7 – Development Tools & Technical Justification

Core Technology Choices - Complete the following

- Programming Language: Use Java and Kotlin for an Android app. If allowed to, use Swift to create an iOS app.
- Framework/Engine (if applicable): Flutter extension in VS Code and use the framework that comes with Flutter to create the Android App. We can also use the .NET MAUI framework for the app too.
- Database (if applicable): Google's Firebase database for storing user data and tracking trends over time and stuff.
- Hosting/Deployment Plan (if applicable): Deploying it to the Android App store.
- Additional Libraries/APIs (if applicable): We can use Google Lens API to search the image and get back results on nutrition. We can also use Google's ML Kit or Open Food Facts API for barcode scanning and then OpenAI's ChatGPT API to search it online and get back results.

Why These Tools Are Appropriate – Write 1-2 well-developed paragraphs that address:

- How these tools support your product requirements
- Why they fit your team's skill level
- Whether they avoid creating a single point of failure
- Any classroom or technical constraints

These tools support our product requirements because they streamline and effectively provide search tools for scanning labels on the back of food products and searching them up too to provide relevant data back. Our team's skill level is appropriate for this project because some of us have had some experience using APIs, but not really with Android development. We have experience with Java from CSA and Data Structures, but not really Kotlin (we think they're really similar). We can combine our knowledge from Java and figure our how to code with Kotlin and code an Android app (it'll be fun)!

I think the authentication with the database is a single point of failure because if the user cannot login, then the user cannot get the API keys and tokens, so the app is pretty much useless for them at that point. They can't have their own local user data and cloud data either. For classroom or technical constraints, some API sites may be blocked on school WIFI and the development SDKs and the frameworks might also be blocked for the downloads. This might prevent us from developing the app. Also, we might need to use an Android phone to test our app out, but that's not allowed in the classroom. So in the class, we will most likely just have a proof of concept.

Section 8 – Known Risks and Assumptions

Technical Risks (at least 2)

- Risk 1: AI hallucinations could happen and it could provide wrong data back for the user. This might make the user angry about the information they receive from the app.
- Mitigation Plan: Provide a feedback button and submit a report for misinformation given by the app, so we can improve the app later with the feedback and train AI models to prevent future misinformation.

- Risk 2: Barcode identification fails or the AI cannot read the information from the image and the user doesn't get the proper data back from the app (the app doesn't know what the barcode corresponds to and gives back blank info).
- Mitigation Plan: Allow the user to type to the AI. Although this may be tedious, the user can communicate with the AI by typing the ingredients, which bypasses the image stuff altogether.

Teamwork Risks (at least 1)

- Risk: Team member has a skill gap and doesn't know what is going on because they don't understand the code and framework.
- Mitigation Plan: The team starts off with the very basics of the app to ensure that the fundamentals are learned first (like UI and API communication) before transmitting actual data and making the UI responsive, etc.

Timeline Risks (at least 1)

- Risk: We don't have a good definition of done and the scope of the project creeps past the definition of done and we start adding new features. Once this happens, we may miss fundamental parts of the project and add other features that we don't need.
- Mitigation Plan: Make a collective decision on if the features being worked on are included in our definition of done and be transparent with our work. This will make it so we don't work on features we don't need while other features that we do need are not done yet.

Section 9 – Success Metrics

How will you measure if your product is successful?

Examples:

- Users can complete core flow
- All MVP features demonstrable
- No major runtime errors
- Performance benchmarks met

List 3-5 success metrics:

- Users can perform the full flow of scanning a barcode/nutrition label, receiving insights on the product and viewing the recommended substitutions/analysis
- No major runtime errors that interfere with the functioning of the app
- Standards regarding speed of analysis/scanning are met
- All MVP features are met